

Principle of Moments

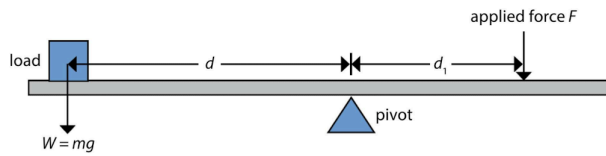


Figure 5.5 The forces acting on both sides of a pivot of a balanced lever

Lever $\Rightarrow$ balanced or in equilibrium

Definition Principle of moments states that when a body is in equilibrium, sum of clockwise moments about a pivot is equal to sum of anticlockwise moments about the SAME pivot

#PhysicsDefinitions

Conditions for Equilibrium

$\Rightarrow$ for body to be stationary or in equilibrium $\Rightarrow$ must not have any rotational or translational motion

$\rightarrow$ no resultant forces or resultant moments acting on the body

Conditions for equilibrium:

- resultant moment on body is 0
- resultant force on body is also 0